SPINAL CORD II

LEARNING OBJECTIVES.

- At the end of learning objectives, students should be able to know:
- Ascending and descending tracts.
- Nuclei of spinal cord.
- Lesions of these tracts.
- Common lesions related to spinal cord.

Principles of Cord Organization

1.Longitudinal Arrangement

Fibers (White Matter) ---------- White Column contain fasciculi

Cell Groups (Gray Matter) ------- Gray Column

2) Transverse Arrangement

Afferent & Efferent Fibers

Crossing (Commissural and Decussating) Fibers

3) Somatotopical Arrangement

3-Nuclie in posterior grey horn:

I) Posterior marginal nucleus: Caps the posterior horne. Receives some of the incoming dorsal root fibres.

II) Subtantia gelatinosa: found at the tip of anterior horne, extends the entire length of spinal cord. Act as relay station for pain and temperature fibres. Concerned with sensory associative mechanism. Give rise to lateral spinothalamic tract.

III) Nucleus proprius: lies subjacent to the substantia gelatinosa throughout the entire extent of cord. It is concerned with sensory associative mechanism.

VI) Nucleus dorsalis or thoracic nucleus: At medial part of base of horn extending from C8 to L3 segments. It is relay nuclear column for reflex or unconscious proprioceptive impulses. Give ise to posterior spinocerebllar tract.

Nuclei of spinal cord

- ***1-Nuclei in anterior grey horn:***
- Anterior horn innervate skeletal muscles. Cells in anterior horn arranged
- in three groups.
- **a) Medial group**: presesnt through out the entire length of spinal cord
- Innervate the axial muscles of body.
- **b) Lateral group**: Present in cervical and lumber enlargements. Supplies
- limb muscles.
- **c) Central group**: present only in upper cervical segments. Represents phrenic
- nerve nucleus and nucleus of spinal root of accessory nerve.

- ***2-Nuclei in lateral horn:***
- **a) Intermediolateral nucleus**: Seen at two levels represents both afferents and
- Efferents
- I) From T1 to L2 give rise to pregangiolonic sympathetic fibers.
- II) From S2 to S4 give rise to pregangiolonic parasympathetic fibres.
- **b) Intermedio-medial**: Mostly internuncial neurons.

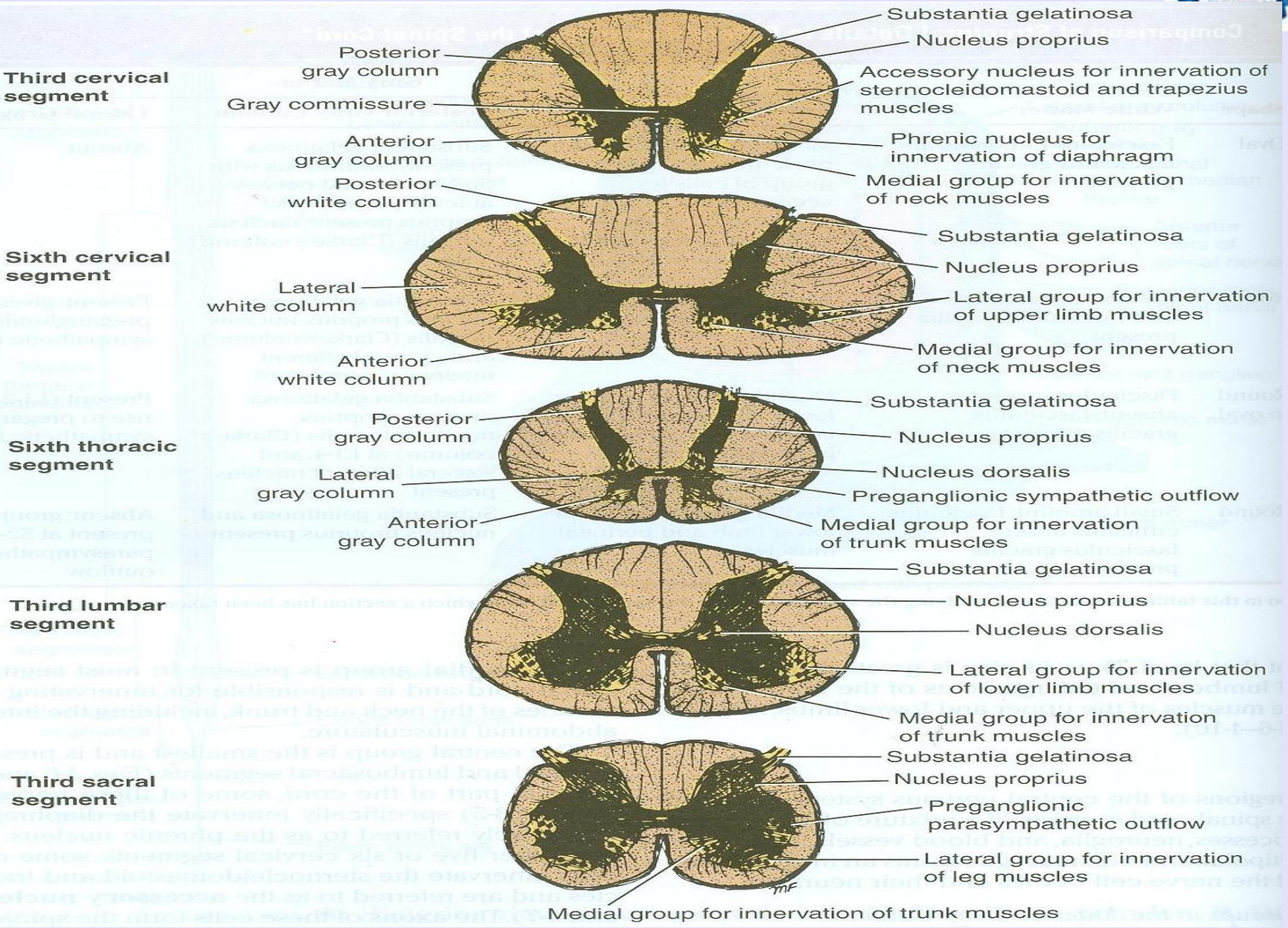

Third cervical segment
Posterior gray column
Gray commissure
Anterior gray column
Posterior white column
Substantia gelatinosa
Nucleus proprius
Accessory nucleus for innervation of sternocleidomastoid and trapezius muscles
Phrenic nucleus for innervation of diaphragm
Medial group for innervation of neck muscles
Sixth cervical segment
Lateral white column
Anterior white column
Substantia gelatinosa
Nucleus proprius
Lateral group for innervation of upper limb muscles
Medial group for innervation of neck muscles
Sixth thoracic segment
Posterior gray column
Lateral gray column
Anterior gray column
Substantia gelatinosa
Nucleus proprius
Nucleus dorsalis
Preganglionic sympathetic outflow
Medial group for innervation of trunk muscles
Third lumbar segment
Substantia gelatinosa
Nucleus proprius
Nucleus dorsalis
Lateral group for innervation of lower limb muscles
Medial group for innervation of trunk muscles
Third sacral segment
Substantia gelatinosa
Nucleus proprius
Preganglionic parasympathetic outflow
Lateral group for innervation of leg muscles
Medial group for innervation of trunk muscles

Table 4-1 Comparison of Structural Details in Different Regions of the Spinal Cord[a]

Region	Shape	White Matter	Gray Matter		
			Anterior Gray Column	Posterior Gray Column	Lateral Gray Column
Cervical	Oval	Fasciculus cuneatus and fasciculus gracilis present	Medial group of cells for neck muscles; central group of cells for accessory nucleus (C1-5) and phrenic nucleus (C3-5); lateral group of cells for upper limb muscles	Substantia gelatinosa present, continuous with Sp.N. of cranial nerve V at level C2; nucleus proprius present; nucleus dorsalis (Clarke's column) absent	Absent
Thoracic	Round	Fasciculus cuneatus (T1-6) and fasciculus gracilis present	Medial group of cells for trunk muscles	Substantia gelatinosa, nucleus proprius, nucleus dorsalis (Clarke's column), and visceral afferent nucleus present	Present; gives rise to preganglionic sympathetic fibers
Lumbar	Round to oval	Fasciculus cuneatus absent; fasciculus gracilis present	Medial group of cells for lower limb muscles; central group of cells for lumbosacral nerve	Substantia gelatinosa, nucleus proprius, nucleus dorsalis (Clarke's column) at L1-4, and visceral afferent nucleus present	Present (L1-2 [3]); gives rise to preganglionic sympathetic fibers
Sacral	Round	Small amount; fasciculus cuneatus absent; fasciculus gracilis present	Medial group of cells for lower limb and perineal muscles	Substantia gelatinosa and nucleus proprius present	Absent; group of cells present at S2-4, for parasympathetic outflow

Ascending tracts

- In posterior white column.

1. Fasciculus gracilis.
2. Fasciculus cunateus. Concerned with.
    1. Light touch.
    2. Vibration.
    3. proprioception

- Tracts in lateral white colomn.

1. Posterior spinocerebellar & anterior spinocerebellar tract. Convey
    1. Proprioceptive information & contribution from touch & pressure receptors.

Ascending tracts in lateral column

- 3. lateral spinothalamic tract concerened with pain & temperature.
- 4. spinotectal tract provide ascending pathway for spinovisual reflexes.
- 5. posterolateral tract.
- 6. spinoreticular mixed with lateral spinothalmic tracts.
- 7. spinoolivary tract. Convey cutaneous & proprioceptive organs.

Descending tracts in anterior white column

1. Lateral corticospinal tracts concerned with voluntary movements.
2. Rubrospinal coney impulse concerned with muscular activity.
3. Lateral reticulospinal tract role in mascular activity.

- 4. desceding autonomic fibres. Control visceral functions.
- 5. olivospinal present only in upper cervical segments associated with muscular activity.

Intersegmental tracts

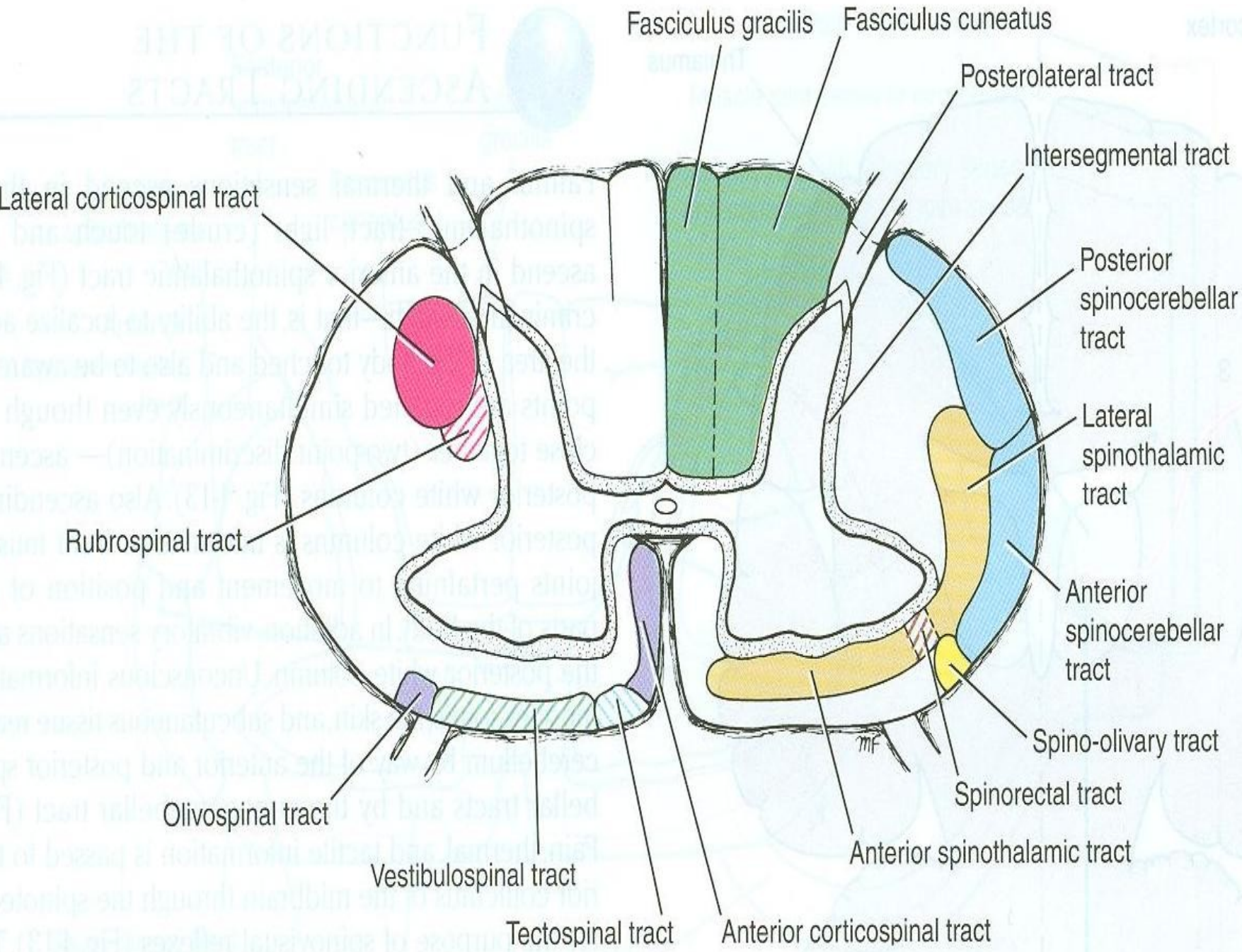
1. in posterior white column.
    1. Posterior intersegmental tract.
2. In lateral white column.
    1. Lateral intersegmental tract contain reticulospinal & descending autonomic fibers.

- 3. tracts in anterior white column.
    - Anterior intersegmental tract.

descending tracts in lateral white column

1. Anterior corticospinal tract confined to cervical & superior half of thoracic region form pathway concerned with voluntary movement.
2. Vestitublospinal convey information of equilibratory controll& muscle tone

- 3.tectospinal tract forms reflex nervous pathway associate with rotation of head & moving arms in response to visual stimuli.
- 4. reticulospinal concerned with motor function.


Fasciculus gracilis
Fasciculus cuneatus
Posterolateral tract
Intersegmental tract
Lateral corticospinal tract
Posterior spinocerebellar tract
Lateral spinothalamic tract
Rubrospinal tract
Anterior spinocerebellar tract
Spino-olivary tract
Spinorectal tract
Olivospinal tract
Anterior spinothalamic tract
Vestibulospinal tract
Tectospinal tract
Anterior corticospinal tract

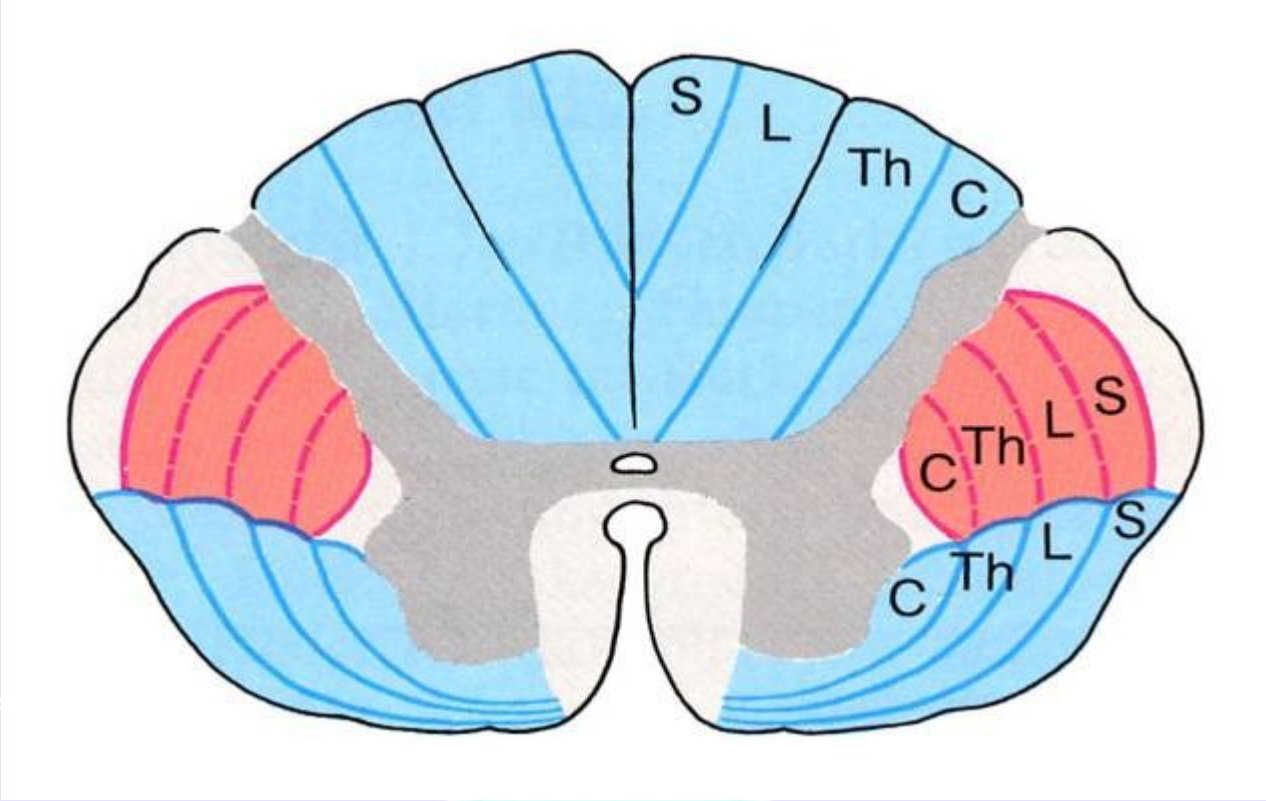
ASCENDING &DESCENDING TRACTS

ASCENDING

- **Fasciculus Gracilis**
- **Fasciculus Cuneatus**
- **Tractus spinothalamicus lateralis**
- **Tractus spinothalamicus anterior**
- **Tractus spinocerebellaris postirior**
- **Tractus spinocerebellaris anterior**
- **Tractus cuneocerebellar**
- **Tractus spinotectalis**
- **Tractus spinoolivaris**

DESCENDING

- **Tractus rubrospinalis**
- **Tractus corticospinalis lateralis**
- **Tractus corticospinalis anterior**
- **Tractus olivospinalis**
- **Tractus tectospinalis**
- **Tractus reticulospinalis**
- **Tractus vestibulospinalis**

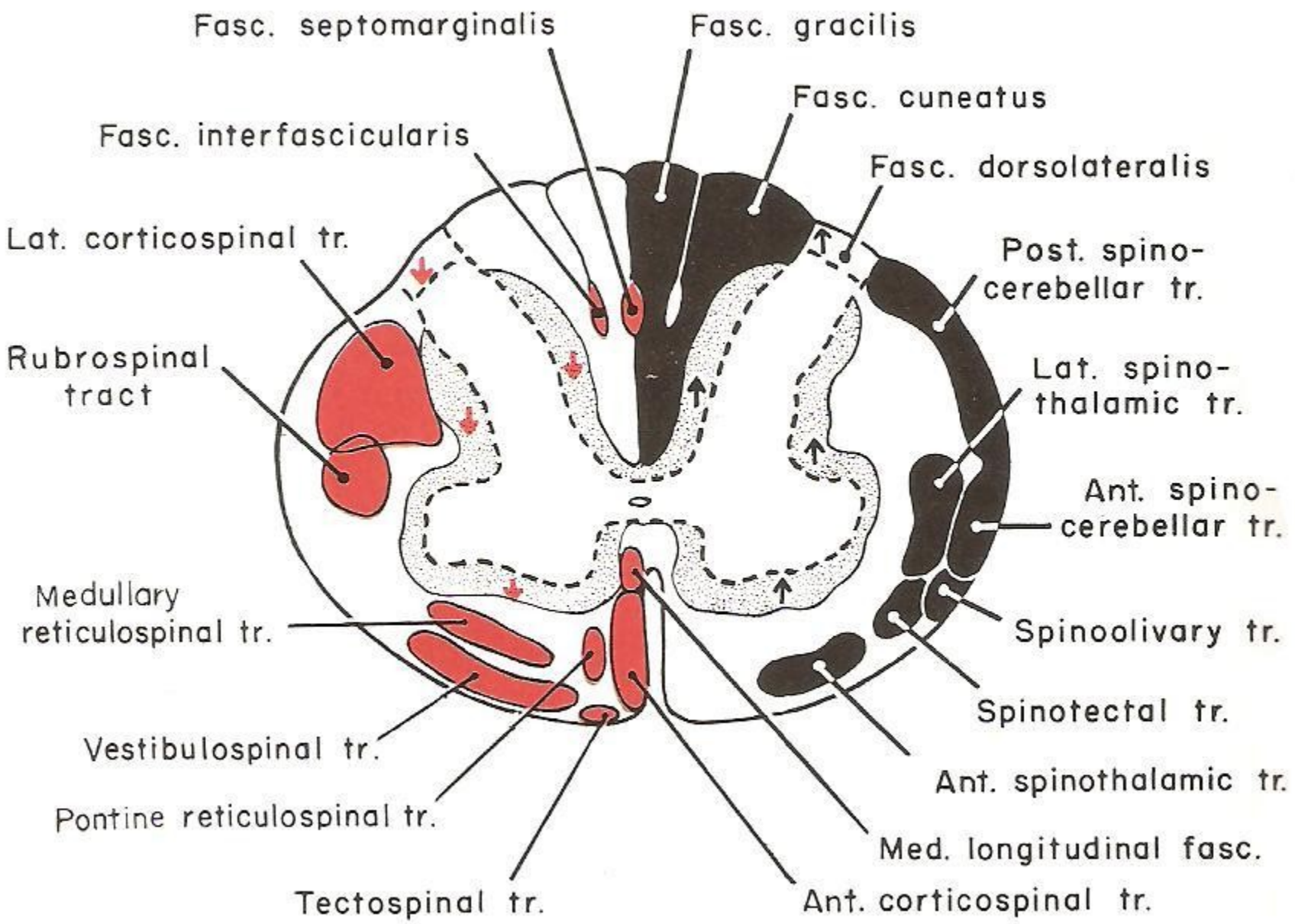


SOMATOTROPIC ARRANGEMENT

TRACTS

Ascending tracts:

- **Anterior spinothalamic---** Light touch and pressure
- **Lateral spinothalamic-----** Pain and thermal sensation
- **F.gracilis &** Two point discrimination, kinesthesis (from muscle and joint),
- **F.cuneatus--** vibaratory sense.
- **Ant: and post:spinocerebellar & cuneacerebllar---** Unconcious muscle, joint, skin and subcutaneous sensations
- **Spinotectal tract---** Pain, thermal and tactile sense to superior colliculus. Spino visual reflex
- **Spinoreticular tract---** From muscle, joint and skin to reticular formation
- **Spino-olivary tract**--- Cutaneous and proprioceptive organs to cerebellum.


Fasc. septomarginalis
Fasc. gracilis
Fasc. cuneatus
Fasc. interfascicularis
Fasc. dorsolateralis
Lat. corticospinal tr.
Post. spino-cerebellar tr.
Rubrospinal tract
Lat. spino-thalamic tr.
Ant. spino-cerebellar tr.
Medullary reticulospinal tr.
Spinoolivary tr.
Spinotectal tr.
Vestibulospinal tr.
Ant. spinothalamic tr.
Pontine reticulospinal tr.
Med. longitudinal fasc.
Tectospinal tr.
Ant. corticospinal tr.

Ascending Tracts

- **Modality:** ***Touch, Pain, Temperature, Kinesthesia***
- **Receptor:** ***Exteroceptor, Interoceptor, Proprioceptor***
- **Primary Neuron**: ***Dorsal Root Ganglion (Spinal Ganglion)***
- **Secondary Neuron**: ***Spinal Cord or Brain Stem***
- **Tertiary Neuron**: ***Thalamus (Ventrobasal Nuclear Complex)***
- **Termination:** ***Cerebral Cortex, Cerebellar Cortex, or Brain Stem***
-

Posterior White Column-Medial Lemniscal Pathway

Modality: *Discriminative Touch Sensation (include Vibration) and Conscious Proprioception (Position Sensation, Kinesthesia)*

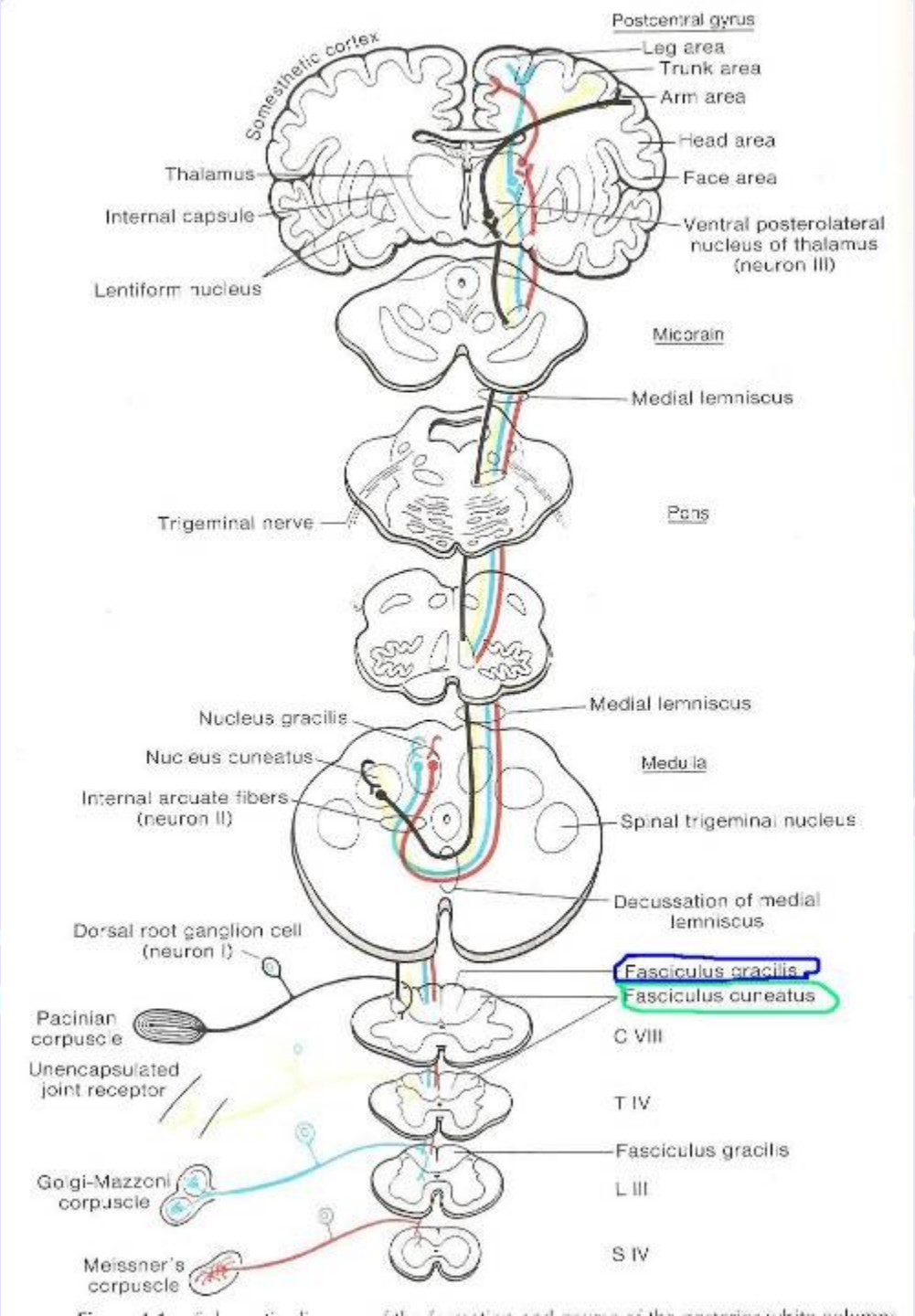
Receptor: Most receptors except free nerve endings

Ist Neuron: *Dorsal Root Ganglion (Spinal Ganglion)*
Posterior Root - Posterior White Column

2nd Neuron: *Dorsal Column Nuclei (Nucleus Gracilis et Cuneatus)*
Internal Arcuate Fiber – sensory Decussation
- Medial Lemniscus

3rd Neuron: *Thalamus (VPLc)*
Internal Capsule ----- Corona Radiata

Termination: *Primary Somesthetic Area (S I)*


Postcentral gyrus
Somesthetic cortex
Leg area
Trunk area
Arm area
Head area
Face area
Thalamus
Internal capsule
Lentiform nucleus
Ventral posterolateral nucleus of thalamus (neuron III)
Midbrain
Medial lemniscus
Trigeminal nerve
Pons
Medial lemniscus
Nucleus gracilis
Nucleus cuneatus
Medulla
Internal arcuate fibers (neuron II)
Spinal trigeminal nucleus
Decussation of medial lemniscus
Dorsal root ganglion cell (neuron I)
Fasciculus gracilis
Fasciculus cuneatus
Pacinian corpuscle
C VIII
Unencapsulated joint receptor
T IV
Fasciculus gracilis
Golgi-Mazzoni corpuscle
L III
Meissner's corpuscle
S IV

LESION CAUSE

- ***ipsilateral***

loss of discriminative touch sensation and conscious proprioception *below* the level of lesion

Spinothalamic Tract:

Modality: ***Pain & Temperature Sensation, Light Touch***

Receptor: ***Free Nerve Ending***

Ist Neuron: ***Dorsal Root Ganglion (Spinal Ganglion) Posterior Root***

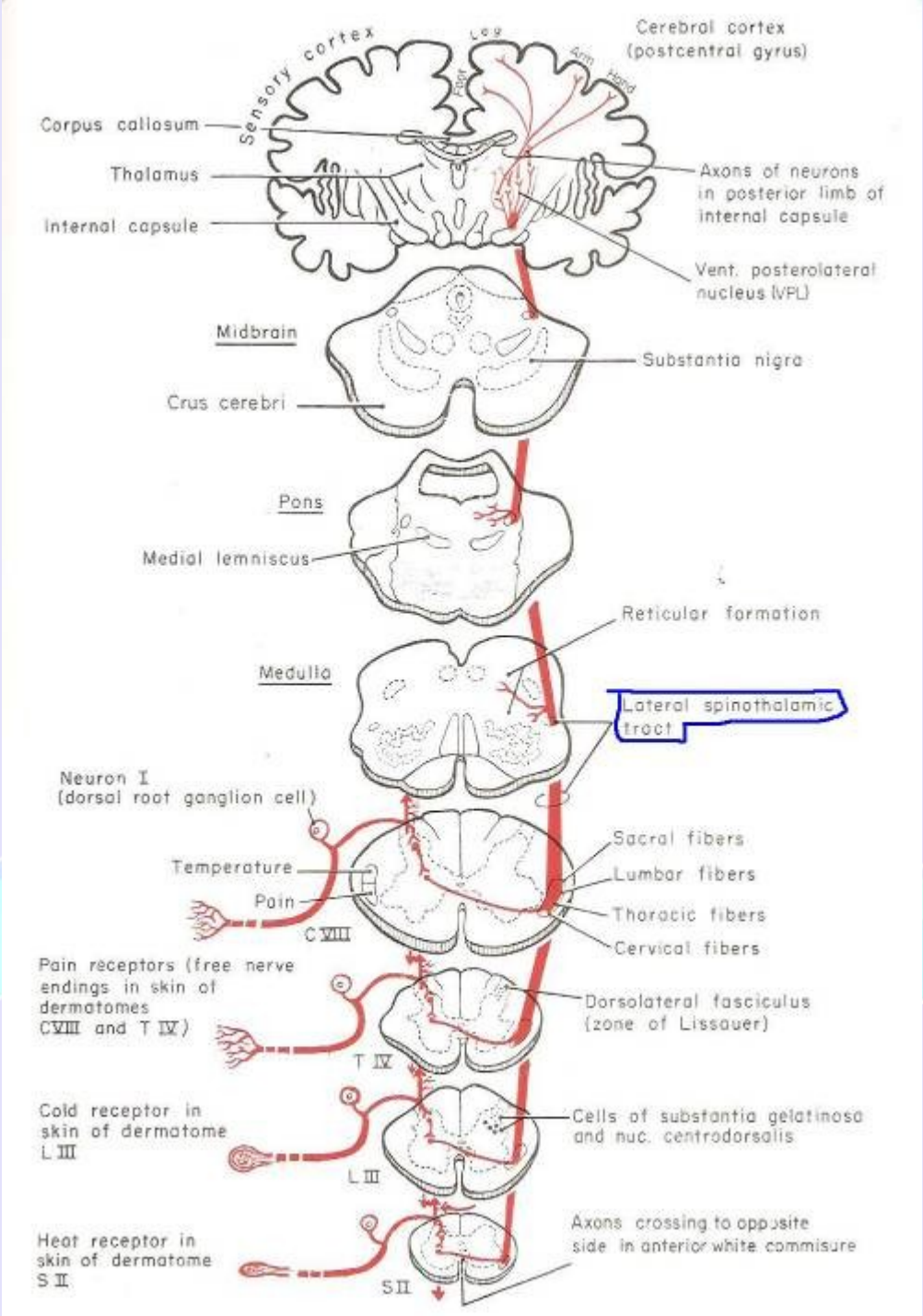
2nd Neuron: ***Dorsal Horn (Lamina IV, V, VI) Spinothalamic Tract - (Spinal Lemniscus)***

3rd Neuron: ***Thalamus (VPLc, CL & POm) Internal Capsule ----- Corona Radiata***

Termination: ***Primary Somesthetic Area (S I) & Diffuse Widespread Cortical Region***

- ***Lateral spinothalamic tract:***
- Pain and temperature. Transmitted in fast conducting delta A type and slow C type fibres.

- Posterior root ganglion → Tip of posterior grey column → Poteriolateral tract of Lissauer → Synapse in posterior column → second order neuron axon cross obliquely to opposite side in ant: commissure ascend as lateral spinocerebellar tract → In medulla joined by ant: spinothalamic and spinotectal forming spinal lemniscus → Third order neuron in VPL Nu of thalamus → Fiber pass through posterior limb of internal capsule and corona radiate to reach → Somesthetic area in postcentral gyrus of cerebral cortex


Cerebral cortex
(postcentral gyrus)
Sensory cortex
Leg
Foot
Arm
Hand
Corpus callosum
Thalamus
Internal capsule
Axons of neurons in posterior limb of internal capsule
Vent. posterolateral nucleus (VPL)
Midbrain
Substantia nigra
Crus cerebri
Pons
Medial lemniscus
Reticular formation
Medulla
Lateral spinothalamic tract
Neuron I
(dorsal root ganglion cell)
Sacral fibers
Lumbar fibers
Thoracic fibers
Cervical fibers
Temperature
Pain
C VIII
Pain receptors (free nerve endings in skin of dermatomes C VIII and T IV)
Dorsolateral fasciculus (zone of Lissauer)
T IV
Cold receptor in skin of dermatome L III
Cells of substantia gelatinosa and nuc. centrodorsalis
L III
Heat receptor in skin of dermatome S II
Axons crossing to opposite side in anterior white commisure
S II

- ***Anterior spinothalamic tract:***

- Light touch and pressure. Like the lateral tract fibres enter the tip of posterior column. Contribute to posteriolateral tract of lissauer. Synaps with cells of substentia gelatinosa → Second order neuron axon cross obliquely in ant: commissure → Joines spinal lemniscus → Same as Lateral spinothlamic.

LESION

Contralateral loss of pain and temperature sensation below the level of lesion

- **Posterior spinocerebellar tract;**

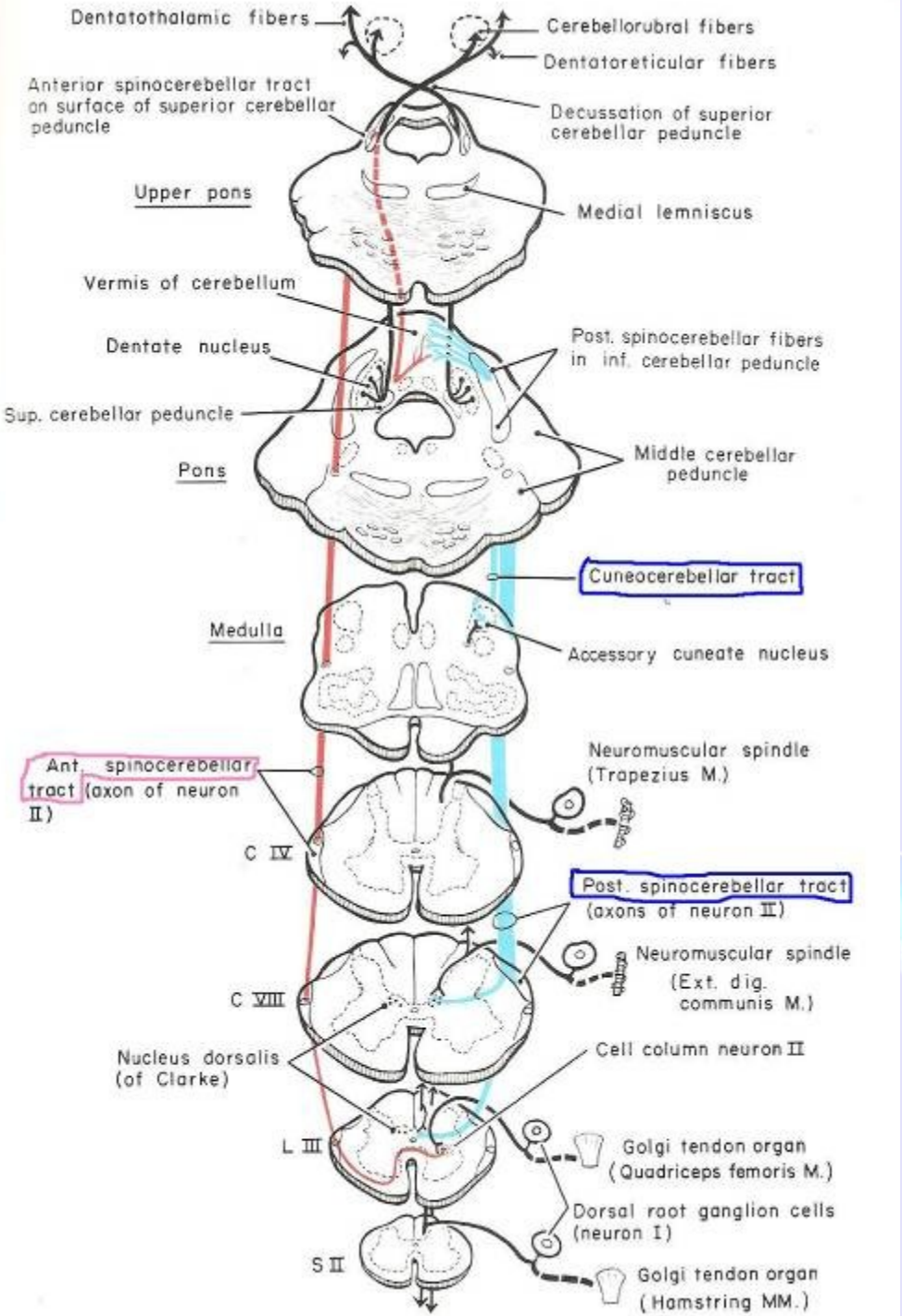
- Muscle and joint sense pathway. Extends from C8 to L3-4 → Fibres from posterior root ganglia synapse in dorsal nucleus of clark's → Axons of second order neuron runs in lateral column → Through the inferior cerebellar peduncle reaches the cerebellum. Golgi tendon, muscle spindle and joint receptor information from trunk and lower limb received by these fibres.

- **Anterior spinocerebellar tract:**

- Concerned with muscle and joint movement information from lower limb. Cells of origin extend from coccygeal to L1 segment. Fibres mostly crossed. Pathway has two neurons. 1st osrder in dorsal root ganglion 2nd in spinal cord. Axons of 2nd order neurons runs over superior cerebellar peduncle to enter the cerebellum. Concerned with coordinated movement and posture of lower limb.

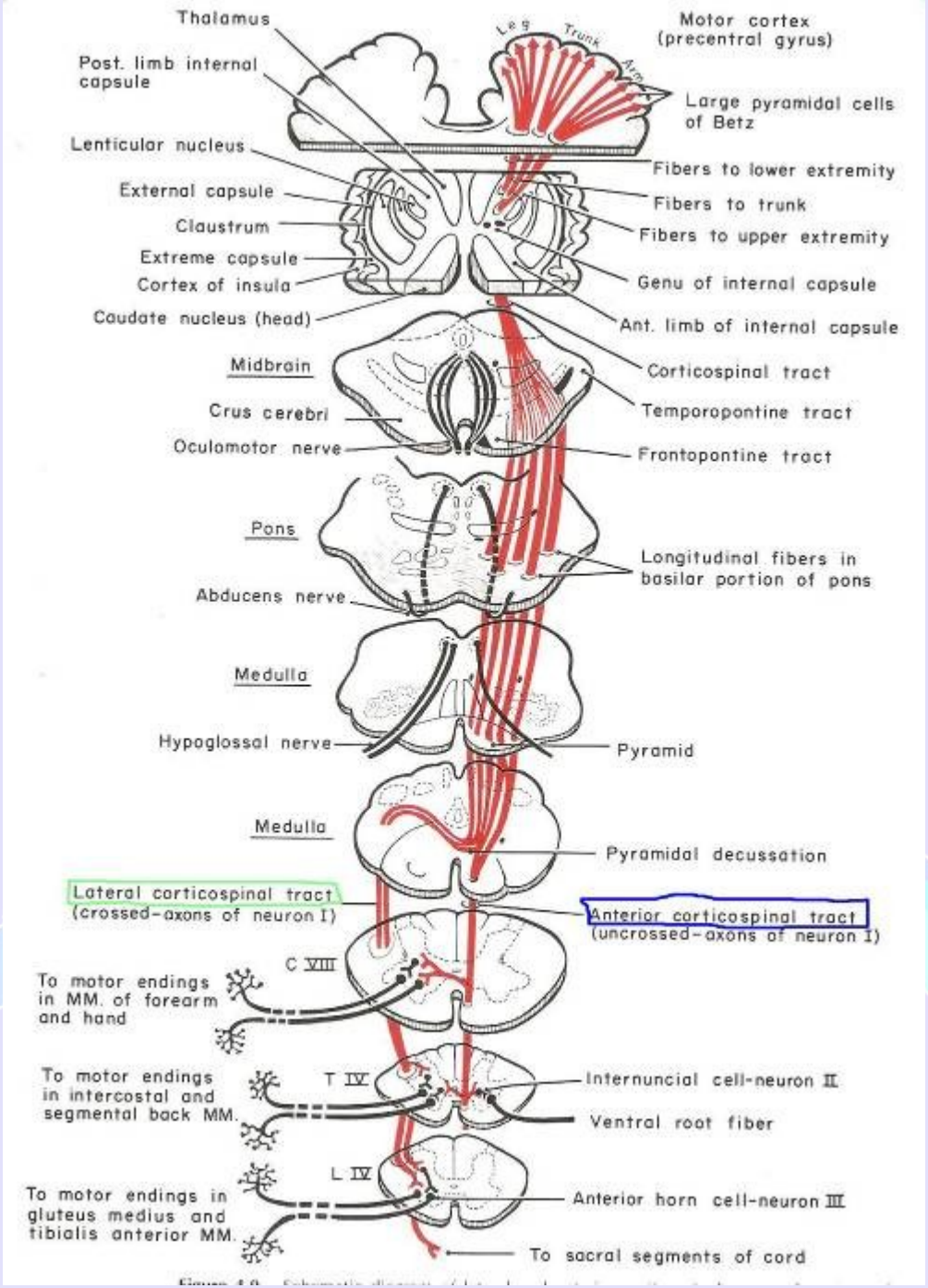
- **Cuneacerebllar tract:**

- Some uncrossed fibres of F.cuneatus synapse on accessory cuneatus nucleus. Equivalent to dorsal nucleus of clark's which is absent above C 8 level. They go cerebellum through inferior cerebellar peduncle. Takes golge tendon and muscle spindle information.


Dentatothalamic fibers
Cerebellorubral fibers
Dentatoreticular fibers
Anterior spinocerebellar tract on surface of superior cerebellar peduncle
Decussation of superior cerebellar peduncle
Upper pons
Medial lemniscus
Vermis of cerebellum
Dentate nucleus
Post. spinocerebellar fibers in inf. cerebellar peduncle
Sup. cerebellar peduncle
Middle cerebellar peduncle
Pons
Cuneocerebellar tract
Medulla
Accessory cuneate nucleus
Neuromuscular spindle (Trapezius M.)
Ant. spinocerebellar tract (axon of neuron II)
C IV
Post. spinocerebellar tract (axons of neuron II)
Neuromuscular spindle (Ext. dig. communis M.)
C VIII
Nucleus dorsalis (of Clarke)
Cell column neuron II
L III
Golgi tendon organ (Quadriceps femoris M.)
Dorsal root ganglion cells (neuron I)
S II
Golgi tendon organ (Hamstring MM.)

DESCENDING TRACTS:

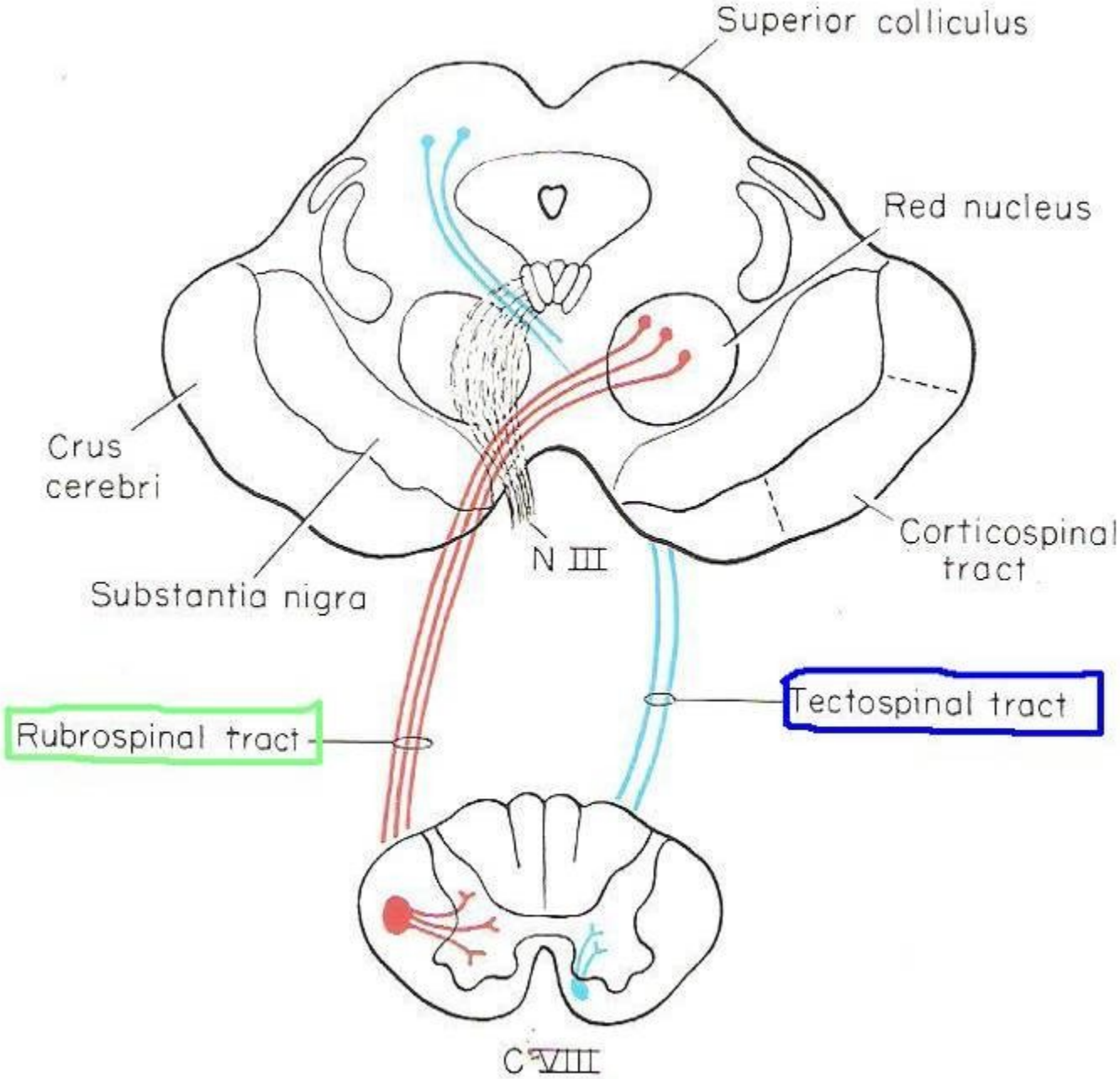
- **Pyramidal tract or corticospinal tract** formed by the axons of pyramidal cells in the motor area of cerebral cortex mostly → Course through posterior limb of internal capsule → Midbrain → Ponse → Medulla here 80% of fibres cross known as pyramidal decussation → Lateral corticospinal tract → Synapse with internuncial neurons in the base of ventral grey column.
- 20% fibres uncrossed form anterior corticospinal tract. They cross at different spinal levels.


Thalamus
Post. limb internal capsule
Lenticular nucleus
External capsule
Claustrum
Extreme capsule
Cortex of insula
Caudate nucleus (head)
Midbrain
Crus cerebri
Oculomotor nerve
Pons
Abducens nerve
Medulla
Hypoglossal nerve
Medulla
Lateral corticospinal tract
(crossed-axons of neuron I)
C VIII
To motor endings in MM. of forearm and hand
T IV
To motor endings in intercostal and segmental back MM.
L IV
To motor endings in gluteus medius and tibialis anterior MM.
Leg
Trunk
Arm
Motor cortex
(precentral gyrus)
Large pyramidal cells of Betz
Fibers to lower extremity
Fibers to trunk
Fibers to upper extremity
Genu of internal capsule
Ant. limb of internal capsule
Corticospinal tract
Temporopontine tract
Frontopontine tract
Longitudinal fibers in basilar portion of pons
Pyramid
Pyramidal decussation
Anterior corticospinal tract
(uncrossed-axons of neuron I)
Internuncial cell-neuron II
Ventral root fiber
Anterior horn cell-neuron III
To sacral segments of cord

- **Extra pyramidal tract**

  a) Ruberospinal tract: formed by the axons of red nucleus cross in the tegmentum and form ventral tegmental decussation. Enters the lateral white column of spinal cord, terminates by synapsing through internuncial neurons with the anterior horn cells.

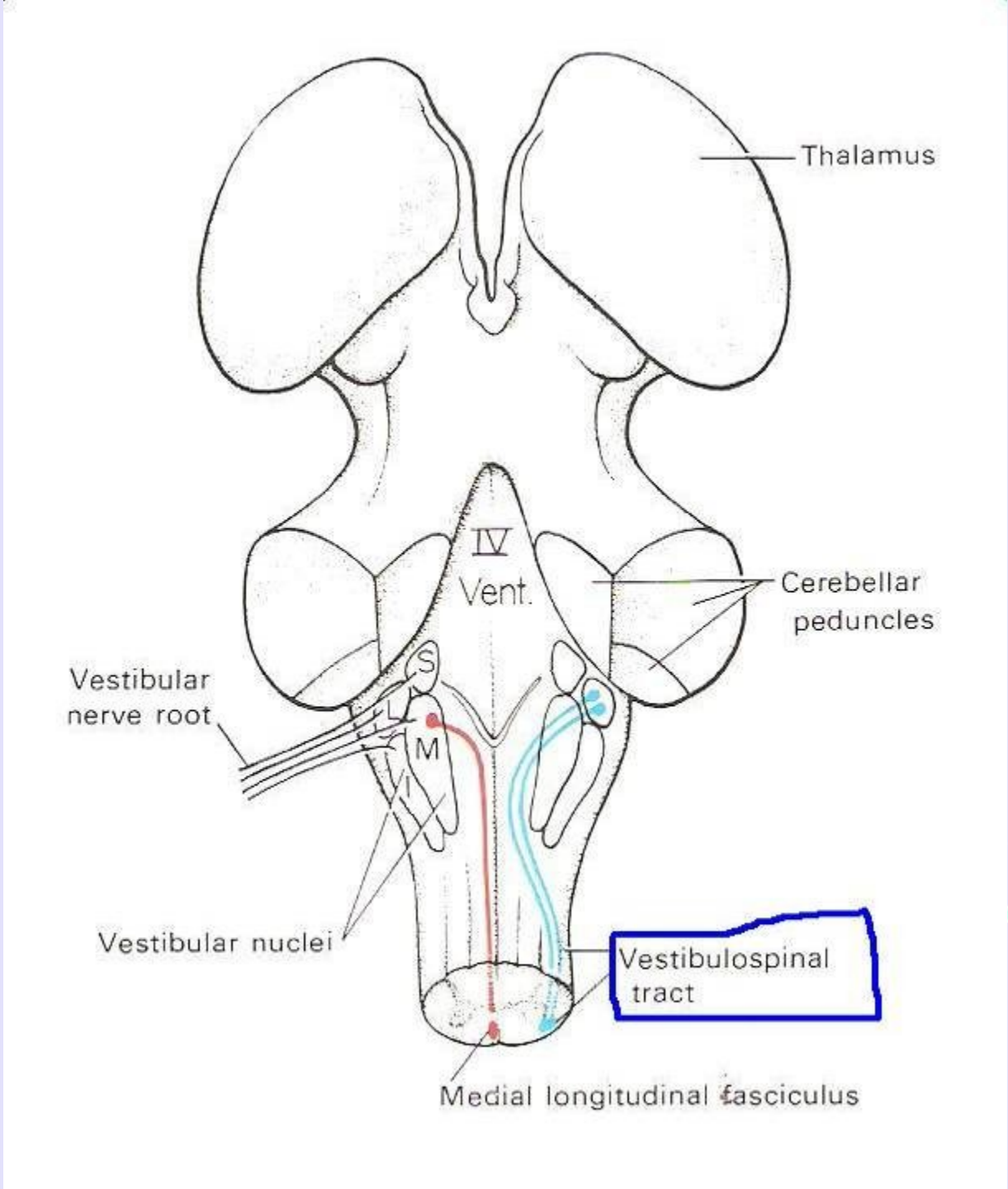
  b) Tectospinal tract: formrd by the axons of neurons in the superior colliculus. Fibres cross to form dorsal tegmental decussation. Fibre terminate on cells of anterior horn cells through internuncial neurons.

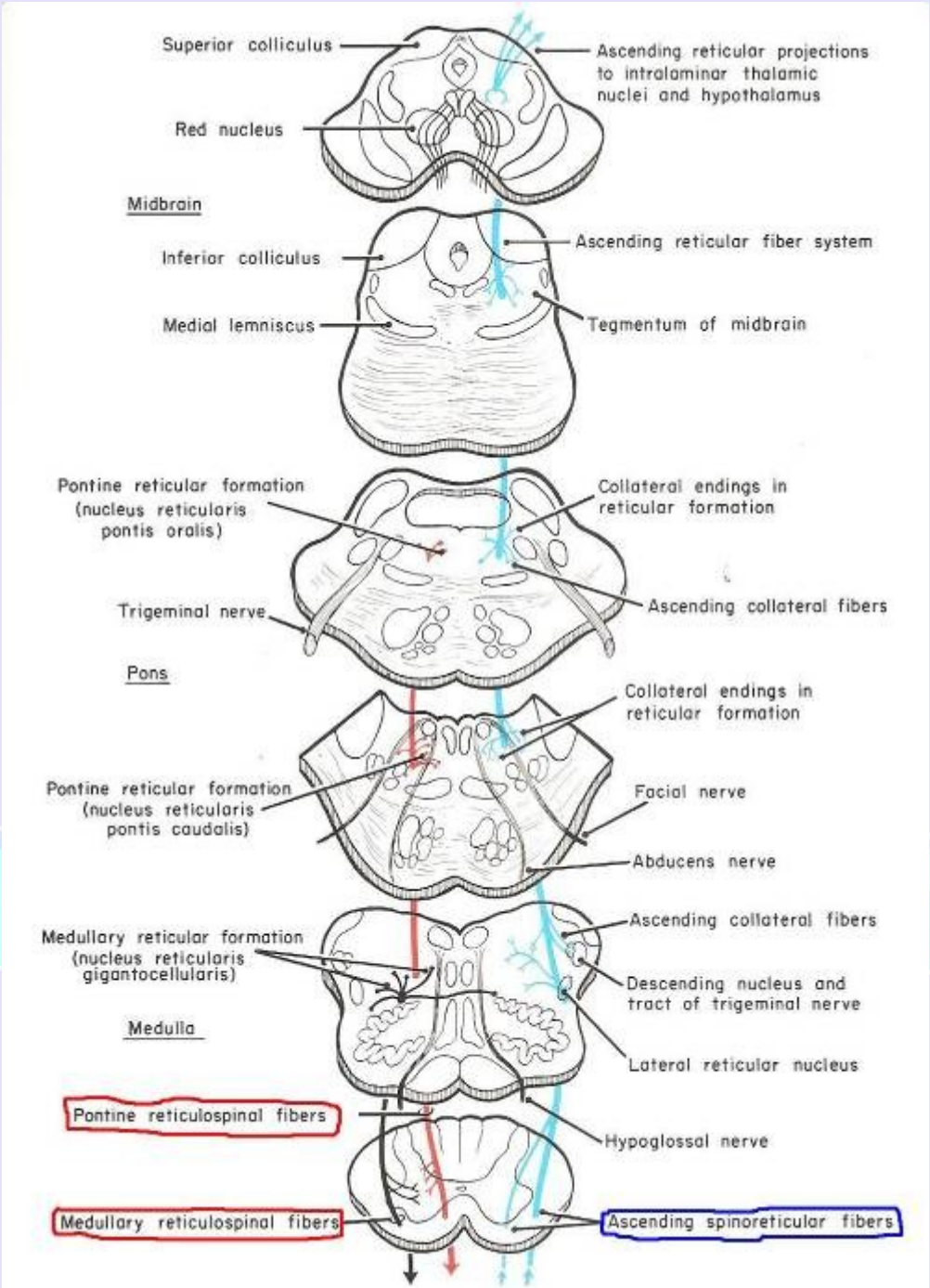

Superior colliculus
Red nucleus
Crus cerebri
Substantia nigra
N III
Corticospinal tract
Tectospinal tract
Rubrospinal tract
C VIII

c) **Vestibulospinal tract:** Fibres arise from lateral vestibular nucleus. Un crossed fibres reach the spinal cord, run in the anterior white column and synapse with anterior horn cells.

d) **Olivospinal tract:** Fibre originate from inferior olivary nucleus descends in spinal cord in anterolateral column of white matter and synapse with the anterior horn cells.

e) **Reticulospinal tract:** Arises from reticular formation in medulla and pons. Crossed and uncrossed. Autonomic fibres from higher levels descends and traverse portions of reticular formation that give rise to the reticulospinal tract.


Thalamus
IV
Vent.
Cerebellar
peduncles
Vestibular
nerve root
S
L
M
I
Vestibular nuclei
Vestibulospinal
tract
Medial longitudinal fasciculus


Superior colliculus
Ascending reticular projections to intralaminar thalamic nuclei and hypothalamus
Red nucleus
Midbrain
Inferior colliculus
Ascending reticular fiber system
Medial lemniscus
Tegmentum of midbrain
Pontine reticular formation (nucleus reticularis pontis oralis)
Collateral endings in reticular formation
Trigeminal nerve
Ascending collateral fibers
Pons
Collateral endings in reticular formation
Pontine reticular formation (nucleus reticularis pontis caudalis)
Facial nerve
Abducens nerve
Ascending collateral fibers
Medullary reticular formation (nucleus reticularis gigantocellularis)
Descending nucleus and tract of trigeminal nerve
Medulla
Lateral reticular nucleus
Pontine reticulospinal fibers
Hypoglossal nerve
Medullary reticulospinal fibers
Ascending spinoreticular fibers

COMMON LESIONS RELATED TO SPINAL CORD

- *Herpes Zoster*

- inflammatory reactions of spinal ganglion

- ***- severe pain on the dermatomes of affected ganglion***

- *Tabes Dorsalis*

- common variety of neurosyphilis

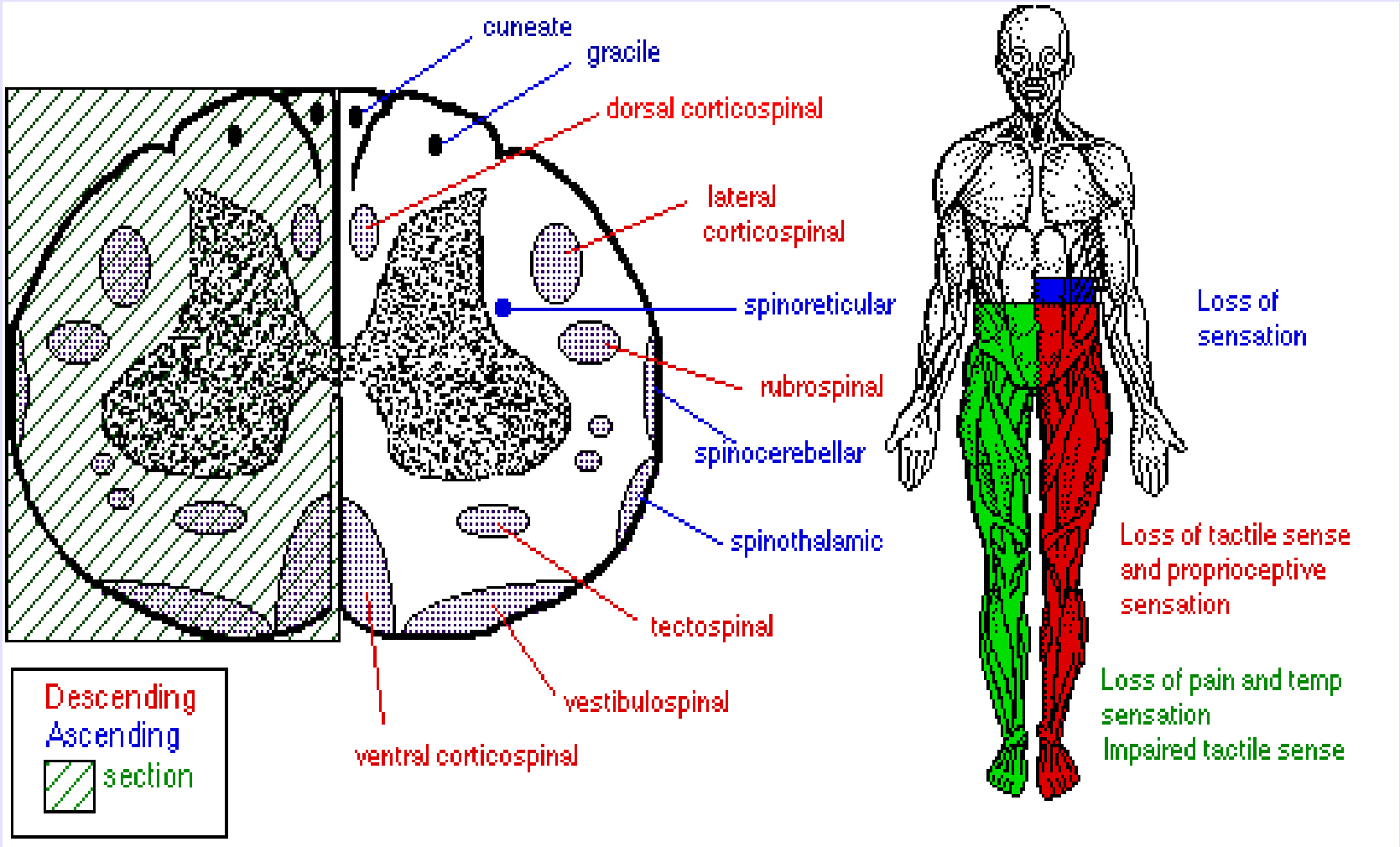
- ***- posterior column and spinal posterior root lesion***
- ***- loss of discriminative touch sensation and conscious***
- ***proprioception below the level of lesion***
- ***- posterior column ataxia***

Brown-Sequard syndrome

- *(spinal cord hemisection)*

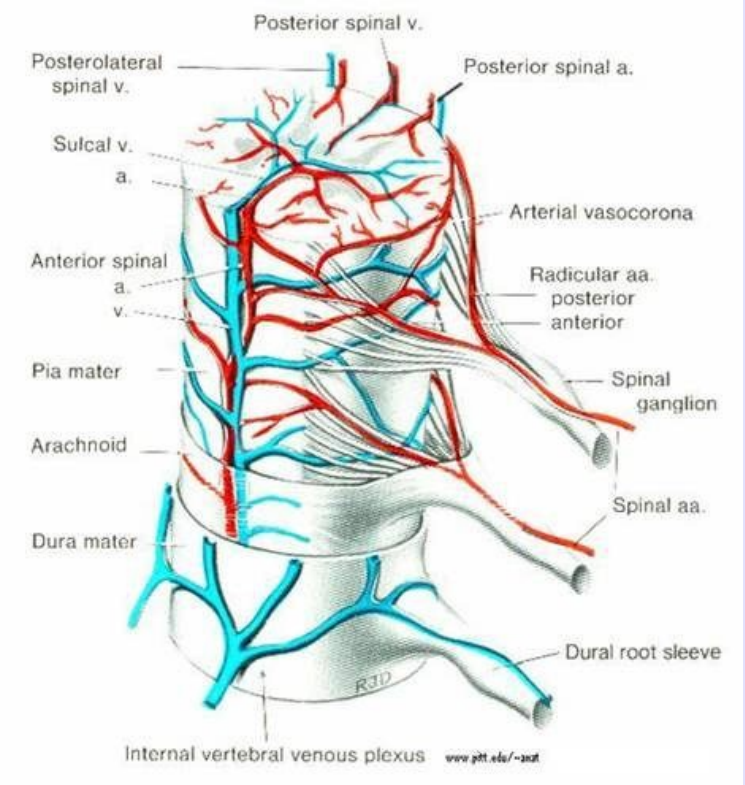
Major Symptoms

- ***1. ipsilateral UMN syndrome below the level of lesion***
- ***2. ipsilateral LMN syndrome at the level of lesion***
- ***3. ipsilateral loss of discriminative touch sensation and conscious proprioception below the level of lesion***
- ***(posterior white column lesion)***
- ***4. contralateral loss of pain and temperature sensation below the level of lesion (spinothalamic tract lesion)***


cuneate
gracile
dorsal corticospinal
lateral corticospinal
spinoreticular
rubrospinal
spinocerebellar
spinothalamic
tectospinal
vestibulospinal
ventral corticospinal
Descending
Ascending
section
Loss of sensation
Loss of tactile sense and proprioceptive sensation
Loss of pain and temp sensation
Impaired tactile sense

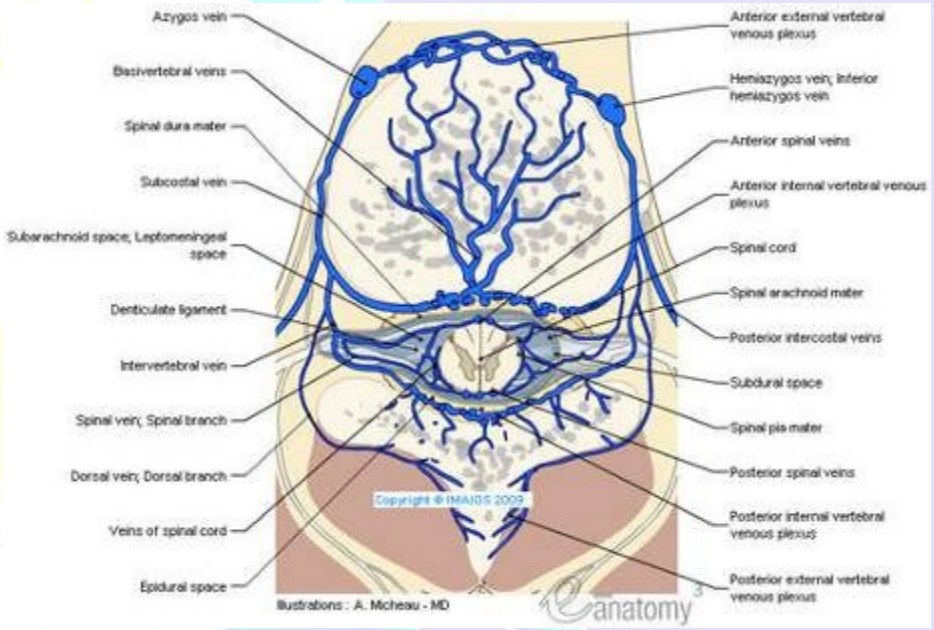
BLOOD SUPPLY

- The spinal cord is supplied by the (single) anterior and (right and left) posterior spinal arteries which descend from the level of the foramen magnum and form three longitudinal channels from which branches enter the cord. They are supplemented at variable levels by anastomoses with a variable number of radicular arteries.



SPINAL VEINS

- **The spinal veins form loose-knit plexuses in which there are an anterior and a posterior midline longitudinal vein, and on each side a pair of longitudinal veins posterior to the anterior and posterior nerve roots. These veins drain to the internal vertebral venous plexus, and thence via the external vertebral venous plexus to the segmental veins: vertebral in the neck; azygos in the thorax; lumbar in the lumbar region; and lateral sacral in the sacral region. At the foramen magnum they communicate with the veins of the medulla.**



THANK YOU